

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 31-400-1
FULLY AUTOMATED CABLE TEST AND FAULT LOCATING VAN	Date: 17-12-2024

EDMS 31-400-1

TECHNICAL SPECIFICATION

FOR

Fully Automated

CABLE TEST AND FAULT LOCATING VAN

Issue: DEC-2024

- This revision contains option items that must be selected by EDC before bidding.

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1. SCOPE

This specification covers the minimum requirements for cable test and fault locating van for M.V. underground cables up to 22 kV distribution networks. The design, engineering, manufacture, testing at the manufacturer's factory & field site, and delivery at the destination shall be supported.

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification shall comply with the latest edition of international standards and shall be designed, manufactured, and tested in accordance to the latest applicable international standards as follows:

Table (1)

Standard No.	Description
IEC60060-3	High-voltage test techniques
IEC60502-2	Power cables with extruded insulation and their accessories
IEEE400-2012	Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems
IEEE400.2-2013	Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)(less than 1 Hz)
IEC60229	Tests on extruded over sheaths with a special protective function
IEEE62.2	Guide for Diagnostic Field Testing of Electric Power Apparatus - Electrical Machinery
EN61243-2	Live working - Voltage detectors - Part 2: Resistive type to be used for voltages of 1 kV to 36 kV A.C
EN61219	Live Working - Earthing or Earthing and Short-Circuiting Equipment Using Lances as a Short-Circuiting Device - Lance Earthing

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IEC60270	High-voltage test techniques - Partial discharge measurements
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3. ENVIRONMENTAL CONDITIONS

The performance of the system shall be guaranteed for the following environmental conditions, any differences in the guaranteed performance shall be clearly set out in the offer:

Table (2)

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.

4. VEHICLE DESCRIPTION:

Required: - Twin Rear Wheel Van, high Roof, equipped with testing Equipment.

4.1 TECHNICAL SPECIFICATIONS:

- MODEL: Latest model.
- PAYLOAD (Kg.): (2500-4000) net payload.
- ENGINE: Diesel- 4 stroke- Electronically controlled direct injection- forced charging water cooling- heavy duty- 4 Cylinder in-line.
- Max. Output Power: 140 HP. At (3000-3500) r.p.m.
- CLUCTH: Frictional, Self-Adjusting Clearance, Torque is greater than Maximum Engine Torque.

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- TRANSMISSION (GEAR BOX): Mechanical Synchronous Harmonic, 6 speeds (forward, reverse), Torque is greater than Engine Torque.
- PWOR STEERING: (L.H.D) Hydraulic power steering.
- SUSPENSION:
 - Front: Independent front suspension (Heavy Duty)
 - Rear: Alloy steel tapered leaf springs (Heavy Duty)
 - Shock absorbers (FRONT/ REAR): Hydraulic double-acting telescopic type.
- DIMENSIONS (mm):
 - Overall length: Max 5910 mm (including rear sill)
 - MAX. Cabin Width: 2000 mm
 - Internal (L/W/H): (3250 -3300) / (1750 -1800) / (1800 -1850) mm
 - Height: 1940 mm from the ground
 - Load Area (m²): (5.5-6) m²
 - Cargo Capacity (m³): (10.25 -11) m³
- Bulkhead between the instruments cabin and the driving cabin with sliding window.
- Rear Door: Two-part doors (Hinged) open 180°, symmetrically with the rear cabin (Height, Width).
- Sliding Side door at right side with sliding window.
- The roof, floor, and sides shall be electrical & thermal insulated.
- The vehicle should support the parking system.
- Provided with AIR Conditioner (A/C):
 - Front (A/C): In the driver's cabin powered by the engine.
 - Rear (A/C): In the instrument cabin works on 220 volts, with a capacity commensurate with loading devices, and operators.

4.2 COMFORT, CONVENIENCE:

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- The test vehicle shall be equipped with the following:
 1. Indoor and outdoor Mirrors.
 2. Cabin Interior lights.
 3. Three-point seat Belts on all seats according to traffic instructions.
 4. Indicator lights for turn signal, main beam, charging current, oil level, coolant level, brake fluid, brake pads, etc.
 5. Front side tighten Benz.
 6. A necessary kit bag for quick maintenance.
 7. Two traffic-approved fire extinguishers powder type “6 kg”.
 8. A hand lift Wrench Suitable for Vehicle weight, a Wheels key, and one Spare Tire.
 9. Radio et.

4.3 TECHNICAL STIPULATIONS

- The vehicle must conform to industry principles and global specifications, taking into account safety & security, and complete comfort requirements.
- The vehicle must meet the requirements and conditions of traffic law and its executive regulations and environmental legislation of A.R.E.
- The vehicle commonly used in A.R.E. has an agent and a center for maintenance & spare parts within Greater Cairo and its suburbs.
- Submit all necessary documents for licensing.

5. TECHNICAL SPECIFICATION OF SUPPLIED EQUIPMENT

- The overall character of the system must be centrally controlled, fully automated, and fully integrated with digital meters.
- The options check table at point 11 should be filled by the tenderer.

5.1 Control and safety unit:

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- The control and safety unit provides optimized safety features and is easy to operate.
- It includes all necessary elements for the operation of the cable test system such as:
 1. Main switch for mains mode.
 2. Incorporated high-voltage instrument selection suited.
 3. Indications of selected instruments and selected phases.
 4. Emergency off button.
 5. Monitoring and displaying safety functions such as back door switch, Mains on-off, emergency stop, and residual voltage indicator.

5.2 Discharge and earthing switch:

- It's used for resistance discharge of the test burning, surge, and cables for max 80 kV automatically when the system is switched off, with visible earthing.
- Max discharge energy ≥ 10000 W/S.

5.3 Main supply unit:

It will include automatic circuit breakers, mains socket outlets, residual current relay and mains relay, and fuses extra.

5.4 Cable drum racks:

The cable drum racks are study designed in steel frame hand-operated and the drums are equipped with lockable brakes, the rack will include the following individual drums:

1. Three high voltage cable drums 80 kV with a length not less than 50 m flexible H.V. coaxial cable (for every drum) including connection, clamps, lockable plug, and strain relief.
2. Mains cable drum with a length not less than 50 m (copper) with mains plug and equipped with a brake.

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3. Earthing cable drum with a length not less than 50 m, CSA 16 mm², highly flexible copper wire, and contact ferrules at 6 m, each, / transparent color.
4. Auxiliary earth cable drum with a length not less than 25 m, CSA. 2.5 mm², highly flexible, red color with tapping clamp, and spike.

5.5 High Voltage (DC) Testing Set:

- The H.V. testing set is used to test and monitor cables up to 22 kV AC. Up to 80 kV DC.
- The set shall be able to measure the leakage current, with a suitable resolution not less than 1μA. with a minimum reading 10 μA.
- The set shall meet at least the following specifications:
 1. Max output voltage D.C ≥ 80 kV.
 2. Rated current D.C ≥ 10 mA.
 3. Short circuits current D.C > 500 mA.
 4. Measurement accuracy $\leq \pm 3$ %.

5.6 Insulation Resistance Test Set:

- Voltage range ≥ 1000 V DC.
- From 0 to 1 Giga ohm through ranges resolution 1 Mega ohm.

5.7 VLF Test (option):

- Rated voltage: ≥ 54 kV RMS.
- Sinusoidal, rectangular, or Cosine rectangular waveform with true leakage current measurements.
- Frequency from 0.01Hz to 1Hz.
- Test capacity to be decided by the options table at point 11.
- The screen should monitor the waveform shape during the test.

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5.8 Partial Discharge and Tan Delta Test (option):

- Rated output voltage 40 kV RMS.
- Output current ≥ 15 mA.
- Enable high power test 0.1Hz, 5micro farad @ 40 kV RMS.
- Partial discharge diagnoses using VLF sine wave or damped AC.
- Fully automated calibration according to cable length.
- Must monitor PD along the cable while applying voltage.
- Ability to save measurements and data with easy browsing.
- A PD-free cable should be attached to the test set.

5.9 Sheath Test Set (option):

- Suitable for sheath testing, sheath test percolation, and sheath fault pinpointing.
- Voltage 0...10 kV.
- Max Current ≥ 50 mA.

5.10 Burn Unit and Fault Conversion:

- The burn unit is used to decrease the impedance of cable faults in medium and low-voltage cables and networks.
- The wave on the fault location unit during the burning process does not appear momentarily, but it shall be appearing naturally and can be stored.
- The minimum specifications of the burning unit:
 1. Burn unit max voltage to be decided by the options table at point 11.
 2. Capable of full power output to one hour working time.
 3. Safety control device and built-in discharging device.

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5.11 Surge Generator Unit:

- The surge generator unit is used for pre-location of high and low resistance as well as intermittent faults in cables, and pinpointing using the acoustic method.
- The minimum specifications are as follows:
 1. Output surge energy to be decided by the options table at point 11.
 2. Single trigger pulse.
 3. Surge frequency 6 to 20 pulses per minute.
 4. Energy must be monitored on screen according to the optimized voltage level.
 5. Surge shots counter for long-term maintenance.
 6. Variable pulse sequence (not less than 5 steps).

5.12 Audio frequency (option):

- Output power ≥ 200 Watt.
- Output frequencies shall be automatically matched with the load.
- With all accessories and receivers to use for:
 1. Cable tracing.
 2. Pinpoint low resistance fault.
 3. Measure cable depth.

5.13 Pinpoint locating set:

- For pinpointing cable faults, a surge wave receiver is used together with the ground microphone and the surge voltage generator.
- The receiver evaluates the time difference between the acoustic signal and the electromagnetic impulse of the shock discharge and displays the distance to the fault and indicates if you reach a fault or astride away.
- Pinpoint consists of:

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1. Search coil.
2. Surge wave receiver.
3. Ground microphone for all types.
4. Headphones with noise protection.
5. Automatic adjustment of values.
6. Reduction and filtering of background noise.
7. Bright, sun-readable display.
8. DB limiter (according to noise and vibration protection laws 84 dB).
9. Distance measurement in Milliseconds or meters/feet.
10. left-right indicator to keep the operator always on top of the cable.
11. High ground stability of the sensor up to 45.
12. Complete set of chargeable batteries (1 set main + 2 set spare) with a charger.

5.14 Cable fault locator equipment:

- This method is used for the pre-location of faults on cables it operates according to the pulses displaced on a scope screen. The reflected pulse is displaced on a scope screen.
- It is a fully computerized time domain reflect meter system, three or single phase design with an integrated transient recorder, storage of the displayed signal, and menu-driven operation.
- It is used to determine all types of faults i.e. low resistance faults, high resistance faults, and intermittent faults.
- TDR Technical data:
 1. Pulse output voltage from 50 to at least 200 V.
 2. Pulse width 20 ns – 30 μ s.
 3. Measuring ranges (at $v/2 = 80$ m/s) 20m – 1000 km.
 4. Real-time sampling rate > 100 MHz.

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5. Resolution 0.5 m (at $v/2 = 80$ m/s).

6. Propagation velocity $v/2$ from 50 to 140 m/ μ s.

7. Color display. Not less than 17".

– Methods of pre-location:

6. Pulse echo method:

- Directly L1, L2, L3.
- Pair comparison L1/L2, L2/L3, L3/L1.
- Difference L1-L2, L2-L3, L3-L1.

7. Decay method.

8. A.R.M (Arc reflection method) by using both:

- Time domain reflect meter and burning unit (Max Voltage to be decided by option table at point 11).
- Time domain reflect meter and surging unit up to 32 kV.

9. I.C.M (Impulse current method).

5.15 Remote application (option)

- Complete control for all pinpointing methods.
- Should display cable route on the map.
- Must start and stop the surge generator.
- Control setting of audio frequency.
- Must be secure connection and high authentication.

5.16 Cable Mapping and database application (option)

- Arranged database software for cable management, reporting, and archiving.
- Import data from GIS or self-drawn.
- Immediate displaying of pre-location results on the map.

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5.17 Cable selection set:

It is used to pick a definite cable out of several cables laid together.

- Generator:
 - Pulse voltage ≥ 50 V.
 - Pulse current 0 – 90 A.
 - Pulse sequence 5 sec.
 - Protection class IP54.
- Universal receiver:
 - Flexible sensor with 250 mm diameter.
 - Must indicate current direction.
 - Adding a twisted field sensor is an advantage.
 - The whole set must be presented with a portable case for protection.

5.18 Power supply:

- 220 V A.C. single phase $\pm 10\%$, 50 Hz from mains or power take-off generator controlled by central control.
- Protection against voltage variation $\pm 10\%$, frequency variation $\pm 5\%$ and automatic reset when the input supply becomes healthy.
- Generator “Engine Driven Alternator”:
 1. The vehicle has a built-in generator.
 2. Engine power supply (power take-off) 220 V, 50 Hz, single phase with output power enough for testing equipment and air condition, lighting, and 5 A accessory loads.
 3. Automatic voltage and speed control.
 4. The engine shall be complete with a silencer, all necessary wiring, a voltmeter, and a frequency meter.

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5. The engine of the alternator shall have an electronic governor to maintain the supply frequency with a tolerance of 3Hz from no load to full load.
6. Voltage regulation shall not exceed 5% from no load to full load.

5.19 Spare parts (option):

- The supplier shall supply spare parts for the maintenance of the van to --EDC as needed for 10 years.
- Spare parts shall be described by the tenderer.

5.20 Operation desk system:

- The operation desk system is used for installation of the control unit with a minimum of 3 racks mounted on the floor.
- A rotating chair with a lock mounted on the floor.

5.21 General requirements:

- Three or single phase system.
- All transformers can be used with different types of insulation, such as solid, oil-sealed, air, resin, or gas.
- All tests shall be done via a control unit without any change in the outer connection.
- The system shall contain a protection device for all equipment and operator safety.
- A separate full control for each compartment (displays I/P and O/P test) so that they don't affect each other in case of any equipment failure.
- Helping users, self-diagnostic of faults, and emergency stops.
- Safety interlocks for protection against error operation.
- Automatically high voltage discharge.

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- Three single cable high voltage with a length of at least 50m on a manual-driven drum with lockable breaks.
- One cable for grounding at least 50m on manual-driven drum with lockable breaks.
- Main cable for external power supply at least 50m on a manual-driven drum with lockable breaks.
- One cable for protective earthing.
- The supplier shall have an improved service center.

6. MAINTENANCE:

The supplier shall render the criteria set for maintenance as follows:

1. ---EDC will do all necessary maintenance from the component replacement level all the way up to the module replacement level.
2. Maintenance aids such as diagnostic flow charts as well as adequate test functions shall be available.
3. The maintenance manual with circuit diagrams and part number to enable the above shall also be available in a well-written form.

7. INSPECTION AND TESTING:

Tests of all components shall be carried out by the manufacturer; a representative from the Distribution Company could attend these tests before acceptance.

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CHECKLIST

DESCRIPTION OF TESTS		Pass(P), Fail (F), Un-completed (U), Not available (NA)	
1	Safety		
2	DC Test		
3	VLF Test		
4	Insulation Test		
5	TDR		
6	Pre-location methods		
7	Pinpointing methods		
8	Burning		
9	Pinpoint Receiver		
10	Audio frequency receiver		
11	Partial discharge and tan delta set		
12	Remote app		
13	Cable book		
14	Vehicle check		

8. TRAINING:

- An engineer shall inspect the test van in the factory.
- Two engineers shall be trained on how to operate and maintain the van in the factory.
- All travel expenses including VISA, hotels, tickets, transportation, and hospitality are supported by the supplier.

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9. CATALOGUES:

- All necessary drawings including layout, dimensions, single-line diagram of primary circuits, and the protection shall be submitted with delivery.
- Technical catalogues for each equipment containing the technical data for all offered components of the unit shall be submitted.
- Technical catalogues for vehicles.
- Maintenance manual for each equipment.

10. SUBMITALES:

- A certificate of origin country shall be provided for all equipment.
- Documents shall be supported with delivery for all equipment inside the van including the following data:
 - General description.
 - Theory of operation.
 - Diagrams and circuits description.
 - Detailed description and interconnection diagrams.
- Result test in factory.
- Technical, operation, maintenance book, and services manuals precisely for all equipment.
- Operation guide.
- Copy of software for the module system.

N.B.

- All documents should be in English.

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11. Options Table:

This table shall be marked by the tenderer.

System type	Single phase		Three phase	
5.7 VLF Test	YES		No	
VLF Testing Test Capacity	$\geq 3 \mu\text{F} @ \geq 38 \text{ kV}$ RMS @0.1 Hz		$\geq 5 \mu\text{F} @ \geq 54 \text{ kV}$ RMS @0.1 Hz	
5.8 Partial discharge and tan delta test	YES		NO	
5.9 Sheath Test Set	YES		NO	
5.10 Burning Unit Max Voltage	$\geq 15 \text{ kV}$		$\geq 20 \text{ kV}$	
5.11 Surge Generator Unit	2 kV $\geq$ 1000 J	YES		NO
	4 kV	@ $\geq 2000 \text{ J}$		@ $\geq 1500 \text{ J}$
	8 kV 16 kV 32 kV	@ $\geq 3000 \text{ J}$		@ $\geq 2000 \text{ J}$

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5.12 Audio frequency	YES	NO
5.13 Pinpoint Location Set Quantity	Piece/s	
5.15 Remote Application	YES	NO
5.16 Cable Mapping and database application	YES	NO
5.19 Spare parts	YES	NO

12. GUARANTEE:

- The supplier shall guarantee the test van with all equipment against all defects arising out of faulty design or workmanship, or of defective materials as follows: -
 1. For the vehicle: 12 months from the delivery date or 100000 km which comes first.
 2. For the equipment: two years, the first one with a bid guarantee, and the second one with a certificate.
- The good quality and manufacturing of the test van shall be guaranteed.
- Defective or poorly assembled parts of the test van and its equipment shall be replaced or repaired promptly and without charge.

13. GUARANTEE TABLES:

- The tenderer shall fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables shall be rejected.

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Guarantee Table (1) For

No	Item	Specif.
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

We guarantee the data given above for the equipment offered.

Signature:

Date: